Course Descriptions: Statistics Courses

This document provides detailed descriptions for all relevant statistics courses completed by **Benjamin Oliver Yick** during his Bachelor of Computing (Hons) in Data Science and Artificial Intelligence at **Nanyang Technological University, Singapore (NTU)**, an institution whose Nanyang Business School holds both AACSB and EQUIS accreditation. It is submitted to the Board of Admission at the **School of Business and Economics, Maastricht University**, as direct evidence of the required statistical and quantitative background for admission to the **Master of International Business**, with a specialization in **Information Management and Business Intelligence**. As requested, this document lists the name of each course, the number of credits received, and a short course description (each within 10 sentences). The six courses detailed herein total 42 ECTS and encompass probability theory, mathematical statistics, applied data analysis, sampling and survey methodology, information analytics, and the application of quantitative models for business decision-making. These courses, drawn from both the School of Physical and Mathematical Sciences and the Nanyang Business School, collectively demonstrate the level of statistics and mathematics evaluated by the Board of Admission and confirm that the applicant's undergraduate program corresponds with the master's program in its emphasis on data management, statistical modeling, and data-driven decision-making.

Overview of statistics courses

Code	Course	AU	ECTS	Semester
BU5401	Management Decision Tools	3.0	6	2024–2025 Semester 1
CS2400	Foundation of Information Analytics	3.0	6	2024–2025 Semester 2
MH2500	Probability and Introduction to Statistics	4.0	8	2022–2023 Semester 1
MH3500	Statistics	4.0	8	2022–2023 Semester 2
MH3511	Data Analysis with Computer	3.0	6	2022–2023 Semester 2
MH4511	Sampling and Survey	4.0	8	2025–2026 Semester 1

Note: ECTS credits are calculated using the conversion rate of 1 NTU Academic Unit (AU) = 2 ECTS. NTU Academic Units include self-study time within their calculation, whereas ECTS credits account for self-study time separately.

Detailed course descriptions

BU5401: Management Decision Tools

Academic Units: 3.0 AU (6 ECTS) | **Semester:** 2024–2025 Semester 1

Course Objectives

This course aims to develop the use of a scientific approach using mathematical methods and computer software to make managerial decisions. Upon successful completion, students should be able to recognize the characteristics of problems for which quantitative analysis may be appropriate, formulate a real-world problem in mathematical terms, select an appropriate quantitative model and apply it to the formulated problem, use computer software to solve the mathematical model, and interpret the results of the quantitative analysis to make management decisions.

Main Topics Treated

Optimization models for business decisions; Simulation; Inventory control; Waiting line management (queueing theory); Forecasting; Decision making under uncertainty.

Workload

39 contact hours over 13 weeks (3 hours per week comprising lectures and tutorials).

Course Materials

Textbook: Anderson, D.R., Sweeney, D.J., Williams, T.A. and Martin, R.K. (2016), An Introduction to Management Science, 14th Edition, Cengage Learning (ISBN−13: 978−1−111−82361−0). Other References: Chin, C.K. (2016), Essential Basics of Probability, Statistics and Analytics, 1st Edition, SoftML (ISBN−13: 978−981−11−0335−3); Taylor, B.W. III (2012), Introduction to Management Science: Global Edition, 11th Edition, Pearson (ISBN−13: 9780273766407); Hillier, F.S. and Hillier, M.S. (2008), Introduction to Management Science: A Modeling and Case Studies Approach with Spreadsheets, 3rd Edition, McGraw−Hill Irwin (ISBN−13: 978−0−07−312903−7).

Source: NTU Nanyang Business School OBTL+ Syllabus

CS2400: Foundation of Information Analytics

Academic Units: 3.0 AU (6 ECTS) | **Semester:** 2024–2025 Semester 2

Course Objectives

This course introduces the statistical foundations of data science and information analytics for handling massive databases. It covers the statistical concepts required for big data analytics and introduces students to statistical tests and statistical modeling.

Main Topics Treated

Importance of Statistics; Descriptive Statistics; Probability; Counting Techniques; Discrete Probability Distributions; Normal Distribution; Confidence Intervals; Hypothesis Testing for One Population; Hypothesis Testing for Two Populations; Correlation and Linear Regression; Non-linear Regression; Application of Statistics in Information Analytics.

Workload

2 hours 40 minutes of lectures plus 1 hour of tutorials per week, over 13 weeks. Total approximately 48 contact hours per semester.

Course Materials

Study Guide, 8th Edition: a free e-book in PDF format compiled from twelve statistics textbooks (provided by the instructor). The New Cambridge Statistical Tables (NCST), 2nd Edition: compilation of statistical tables for hypothesis tests. Reference Books: Agresti, A. and Franklin, C. (2013), Statistics: The Art and Science of Learning from Data, 3rd Edition, Pearson Education; Johnson, R. and Kuby, P. (2012), Elementary Statistics, 11th Edition, Brooks/Cole; Utts, J.M. and Heckard, R.F. (2015), Mind on Statistics, 5th Edition, Cengage Learning; Montgomery, D.C., Runger, G.C. and Hubele, N.F. (2001), Engineering Statistics, 2nd Edition, John Wiley and Sons.

Source: NTU WKWSCI Course Outline (AY2024/25)

MH2500: Probability and Introduction to Statistics

Academic Units: 4.0 AU (8 ECTS) | **Semester:** 2022–2023 Semester 1

Course Objectives

This is a core mathematical course aiming to develop understanding of fundamental concepts in probability such as random variables, independence, basic probability distributions, conditional expectations and conditional variances, the law of large numbers, and the central limit theorem with applications. The course also prepares students for further courses in probability and statistics (MH3500, MH3512). Upon completion, students will be able to: calculate probabilities of events concerning discrete distributions by counting; calculate conditional probabilities with Bayes' Theorem; describe probability distributions using CDF/PDF; identify appropriate probability distributions for modeling; calculate expectation, variance, MGF, and quantiles; calculate distributions of functions of random variables and covariance; prove or disprove independence of random variables; calculate conditional expectations and variances; and apply the central limit theorem.

Main Topics Treated

Events, probabilities, law of total probability, Bayes' theorem; Independence of events; Discrete distributions (binomial, hypergeometric, Poisson); Continuous distributions (normal, exponential) and densities; Joint distribution, marginal and conditional distributions for discrete and continuous variables; Functions of two or more random variables, order statistics; Expectation and variance; Covariance and correlation coefficient; Markov and Chebyshev inequalities; Conditional expectations, conditional variances, and moment generating functions; Law of large numbers; Central limit theorem with applications.

Workload

3 hours of lectures plus 1 hour of tutorials per week, over 13 weeks. Total 51 contact hours per semester.

Course Materials

Ross, S. (2020), A First Course in Probability, 10th Edition, Pearson (ISBN: 978–0134753119).

Source: NTU SPMS OBTL Document

MH3500: Statistics

Academic Units: 4.0 AU (8 ECTS) | **Semester:** 2022–2023 Semester 2

Course Objectives

This course develops understanding of the statistical concepts of parameter estimation and hypothesis testing that are fundamental for real-life applications of statistics. Upon completion, students will be able to: apply basic probability concepts (PMF, PDF, CDF, expected values, variance, moments) in a statistical context; use standard probability distributions to model statistical scenarios; explain the relevance of the Central Limit Theorem for statistics; construct parameter estimators using maximum likelihood and method of moments; rigorously assess the quality of estimators; analyze asymptotic properties of estimators; construct exact and approximate confidence intervals; explain the purpose and philosophy of hypothesis testing and the meaning of p-values; create and apply hypothesis tests; compute the size and power of a test; and construct most powerful tests using the Neyman-Pearson Lemma.

Main Topics Treated

Review of probability; Random samples, sample mean and sample variance; Distributions derived from the normal distribution; Central Limit Theorem and its significance for statistics; Introduction to parameter estimation and quality criteria for estimators; Constructing good estimators: method of moments and maximum likelihood method; Asymptotic properties of estimators, Cramér-Rao bound and efficient estimators; Confidence intervals for estimators; Introduction to hypothesis testing and Fisher-type tests; Neyman-Pearson tests and Neyman-Pearson Lemma.

Workload

3 hours of lectures plus 1 hour of tutorials per week, over 13 weeks. Total 52 contact hours per semester.

Course Materials

1. Rice, J.A., Mathematical Statistics and Data Analysis, 3rd Edition (ISBN–13: 978–8131519547). 2. Hogg, R.V., McKean, J.W. and Craig, A.T., Introduction to Mathematical Statistics, 8th Edition, Pearson. 3. Casella, G. and Berger, R.L., Statistical Inference, 2nd Edition, Thomson Press.

Source: NTU SPMS OBTL Document

MH3511: Data Analysis with Computer

Academic Units: 3.0 AU (6 ECTS) | **Semester:** 2022–2023 Semester 2

Course Objectives

This course provides basic concepts for data analysis using the R programming language. Students will learn the skills of plotting, summarizing, making inferences, and presenting various types of data. Upon completion, students will be able to evaluate mathematical functions using R, distinguish between measurement scales, explain statistical quantities, construct various plots, perform statistical inference, understand error types in hypothesis testing, analyze categorical data, use parametric methods, and perform linear regression.

Main Topics Treated

Basics of R Programming (syntax, mathematical expressions, variables, vectors, matrices, dataframes, importing and subsetting data, loops); Describing Data (mean, median, standard deviation, variance, inter-quartile range, boxplot, histogram, stem-leaf plot, normality checks, QQ-plot, outliers, transformation); Statistical Inference (sampling distribution, Central Limit Theorem, confidence intervals, hypothesis testing, Type I and Type II errors, p-values); Categorical Data (proportion estimation, testing of proportion parameter, goodness-of-fit test, two-way contingency table, paired contingency table); Multiple Samples (two independent samples, inference on mean difference, two dependent samples, ANOVA test, multiple dependent samples); Nonparametric Tests (quantile test, Wilcoxon rank-sum test, Kruskal-Wallis test, sign test, Wilcoxon signed-rank test, Friedman test); Correlation and Regression (correlation coefficient, confidence intervals, simple linear regression model, inference on parameters, prediction inference, model checking).

Workload

Approximately 50 contact hours over 13 weeks, comprising lectures and computer laboratory sessions.

Course Materials

1. Hothorn, T. and Everitt, B.S. (2014), A Handbook of Statistical Analysis Using R, 3rd Edition, CRC Press (ISBN−10: 1482204584; ISBN−13: 978−1482204582). 2. Crawley, M.J. (2005), Statistics: An Introduction Using R, Wiley (ISBN−10: 0470022981; ISBN−13: 978−0470022986).

Source: NTU SPMS OBTL Document

MH4511: Sampling and Survey

Academic Units: 4.0 AU (8 ECTS) | **Semester:** 2025–2026 Semester 1

Course Objectives

This course gives an introduction to the statistical aspects of taking and analyzing a sample. Students will learn to determine the appropriate sampling design in various situations, use the correct method for analysis, and interpret the results. Upon completion, students will be able to: recognize and describe various sampling designs; compute estimates for population mean, proportion, and total; construct confidence intervals for population parameters; apply ratio and regression estimations to improve accuracy of estimates; determine the required sample size and its allocation under given conditions; and explain the importance of nonresponse and apply techniques to reduce nonresponse rates.

Main Topics Treated

Probability Sampling (types of probability samples, simple random sampling, estimation of population mean/proportion/total, sample size estimation, systematic sampling); Stratified Sampling (theory, sampling weights, estimation, allocating observations to strata, sample size estimation, defining strata, post-stratification); Ratio and Regression Estimations (ratio estimation, regression estimation, selecting sample size, relative efficiency of estimators); Cluster Sampling (one-stage and two-stage cluster sampling, estimation, selecting sample size, cluster sampling with probability proportional to size); Sampling with Unequal Probabilities (one-stage and two-stage sampling with replacement, unequal-probability sampling without replacement); Nonresponse (effects of ignoring nonresponse, callbacks and two-phase sampling, weighting methods for nonresponse, imputation).

Workload

Total 52 contact hours over 13 weeks, consisting of lectures, tutorials, and assignments.

Course Materials

1. Lohr, S.L. (2010), Sampling: Design and Analysis, 2nd Edition, Brooks/Cole (ISBN: 978−0495105275). 2. Scheaffer, R.L. et al. (2012), Elementary Survey Sampling, 7th Edition, Brooks/Cole (ISBN: 978−0840053619).

Source: NTU SPMS OBTL Document